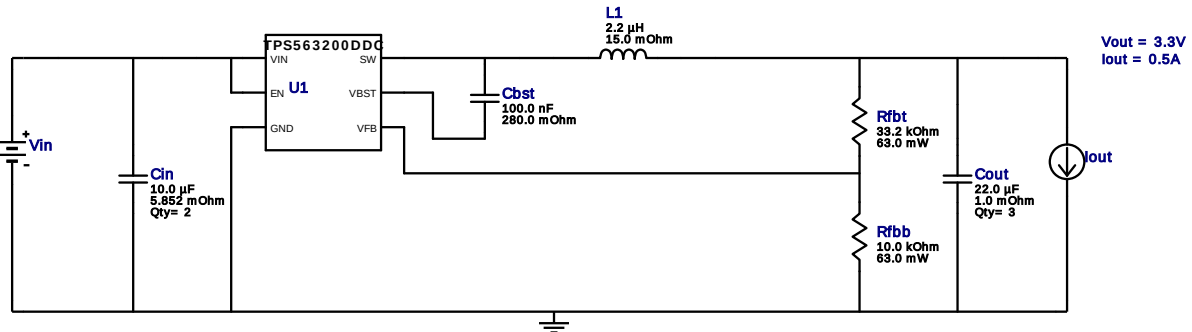


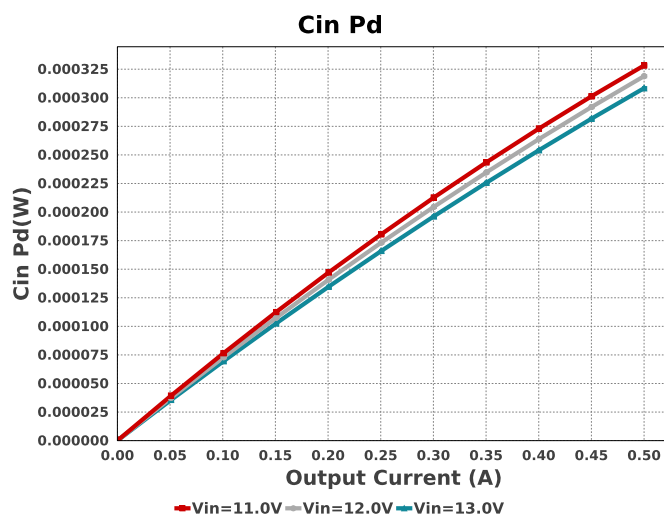
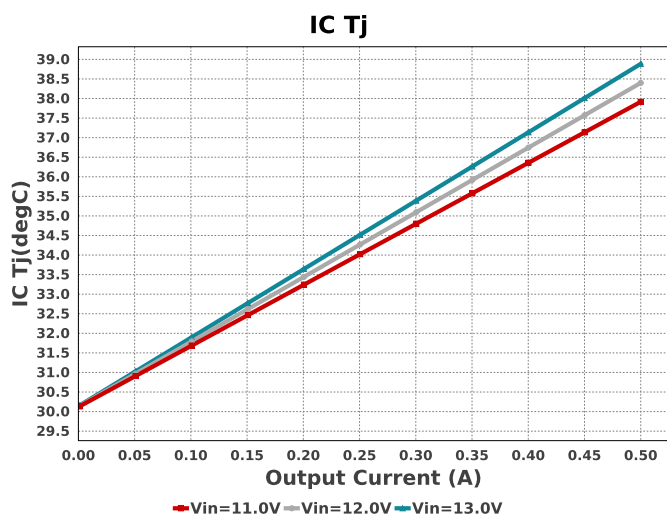
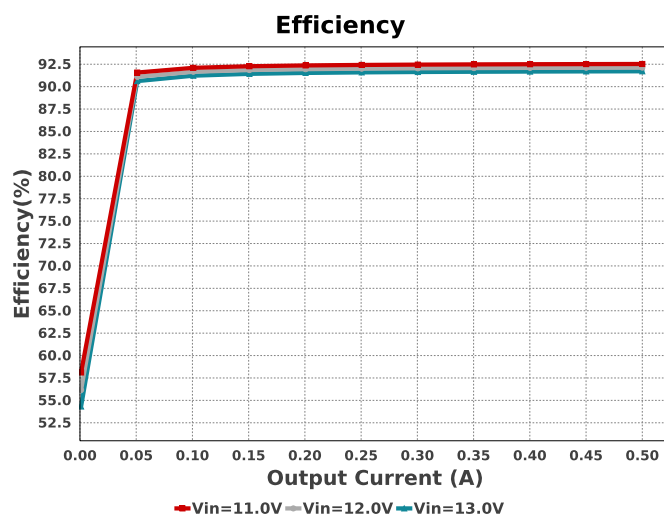
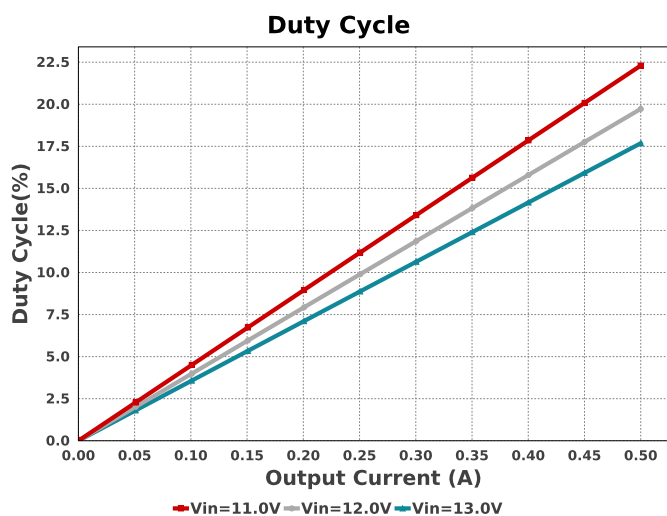
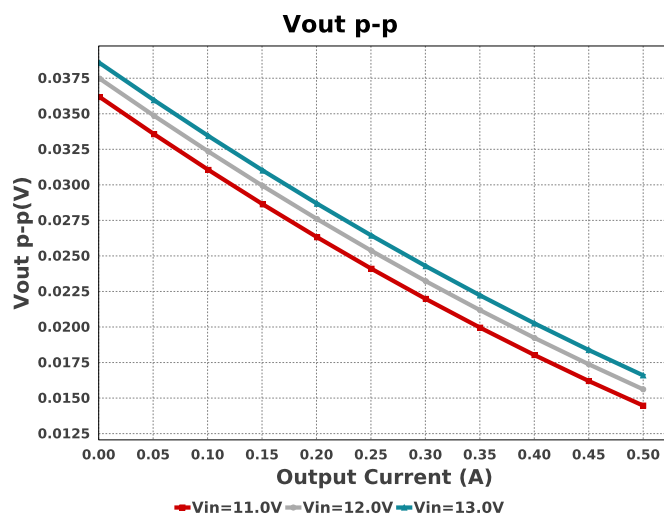
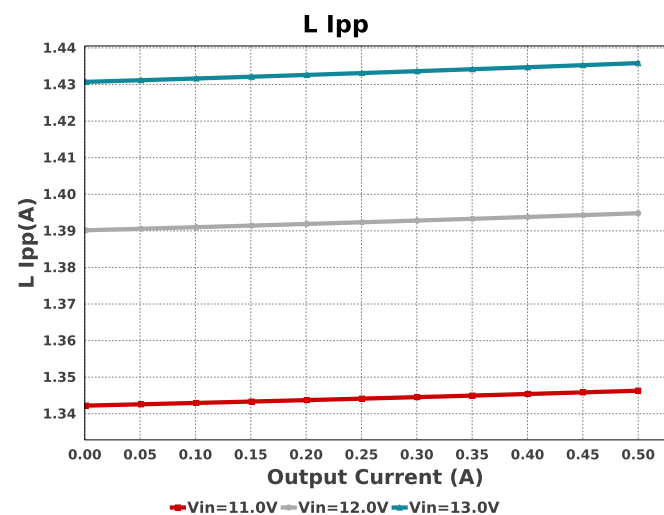
WEBENCH® Design Report

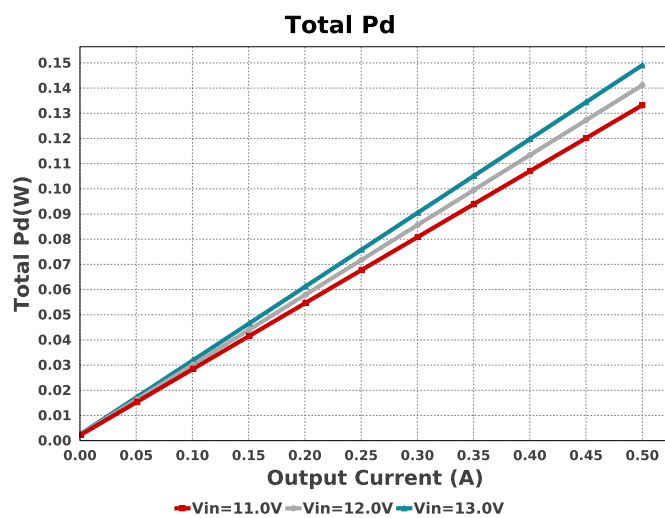
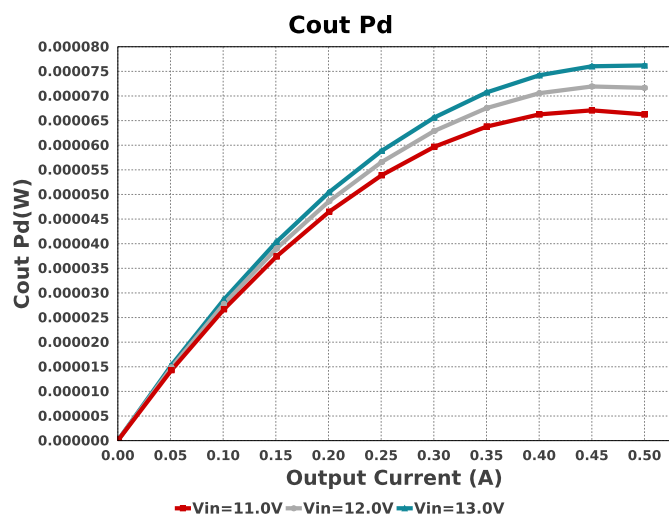
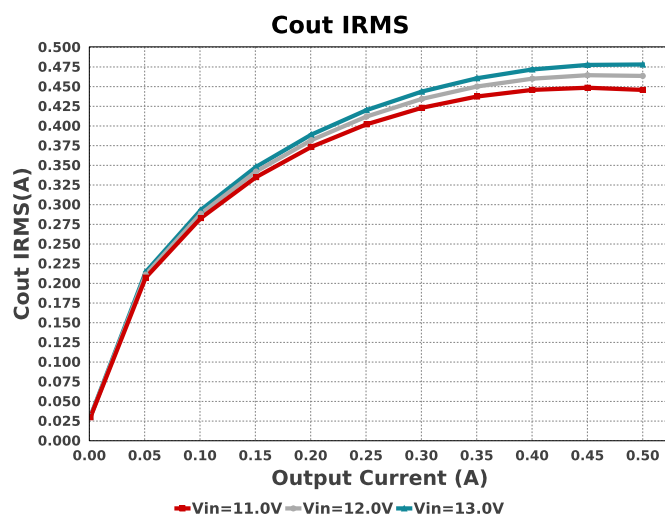
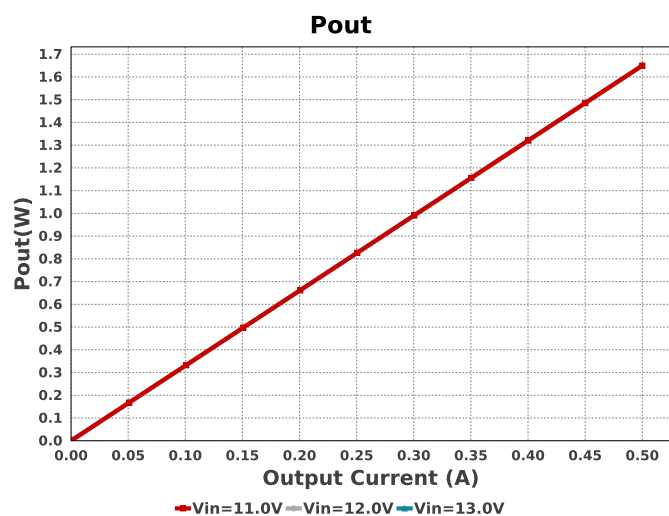
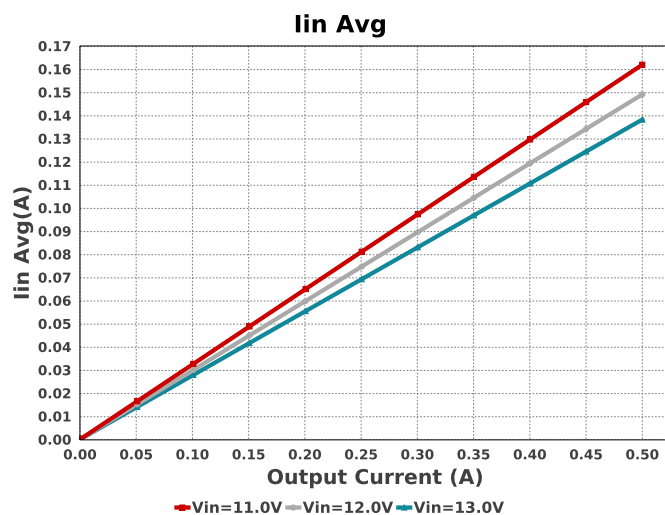
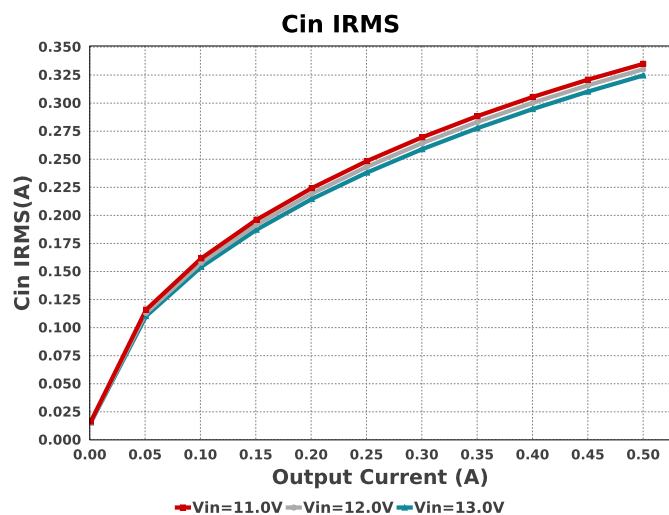
Design : 41 TPS563200DDCR
TPS563200DDCR 11V-13V to 3.30V @ 0.5A

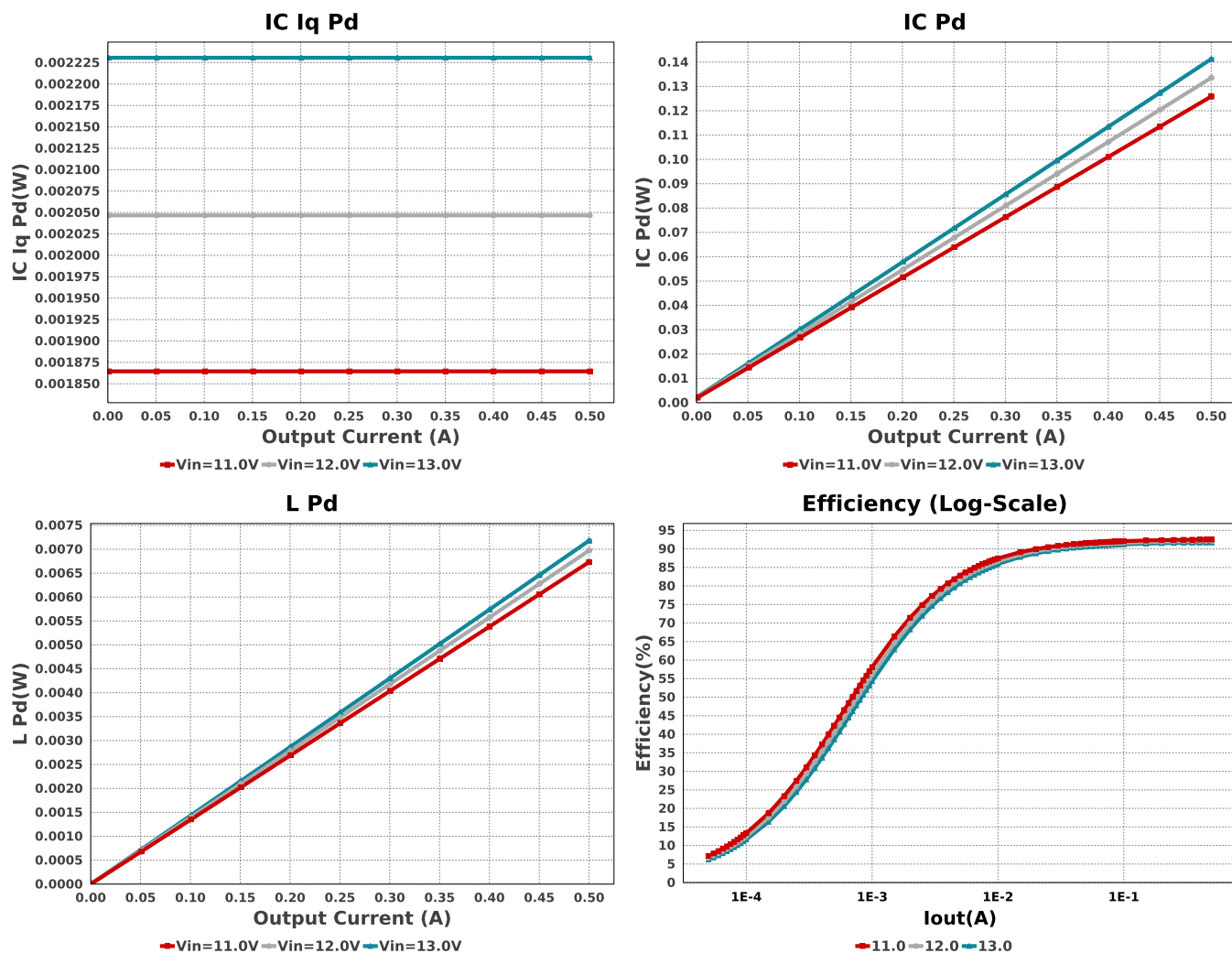


Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cbst	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	\$0.01	0805 7 mm ²
Cin	Taiyo Yuden	MSAST219LB5106KTNA01 Series= X5R	Cap= 10.0 uF ESR= 5.852 mOhm VDC= 25.0 V IRMS= 2.14545 A	2	\$0.09	0805 7 mm ²
Cout	MuRata	GRM188R60J226MEA0D Series= X5R	Cap= 22.0 uF ESR= 1.0 mOhm VDC= 6.3 V IRMS= 6.0 A	3	\$0.04	0603 5 mm ²
L1	TDK	VLP8040T-2R2N	L= 2.2 µH 15.0 mOhm	1	\$0.22	VLP8040 113 mm ²
Rfbb	Vishay-Dale	CRCW040210K0FKED Series= CRCW..e3	Res= 10.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rfht	Vishay-Dale	CRCW040233K2FKED Series= CRCW..e3	Res= 33.2 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
U1	Texas Instruments	TPS563200DDCR	Switcher	1	\$0.34	DDC0006A 10 mm ²







Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	324.642 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	308.38 μ W	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	478.141 mA	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	76.206 μ W	Capacitor	Output capacitor power dissipation
5.	IC Iq Pd	2.231 mW	IC	IC Iq Pd
6.	IC Pd	141.27 mW	IC	IC power dissipation
7.	IC Tj	38.886 degC	IC	IC junction temperature
8.	ICThetaJA	62.9 degC/W	IC	IC junction-to-ambient thermal resistance
9.	Iin Avg	138.39 mA	IC	Average input current
10.	L Ipp	1.436 A	Inductor	Peak-to-peak inductor ripple current
11.	L Pd	7.179 mW	Inductor	Inductor power dissipation
12.	Cin Pd	308.38 μ W	Power	Input capacitor power dissipation
13.	Cout Pd	76.206 μ W	Power	Output capacitor power dissipation
14.	IC Pd	141.27 mW	Power	IC power dissipation
15.	L Pd	7.179 mW	Power	Inductor power dissipation
16.	Total Pd	149.089 mW	Power	Total Power Dissipation
17.	BOM Count	10	System	Total Design BOM count
18.	Duty Cycle	17.68 %	System	Duty cycle
19.	Efficiency	91.713 %	System	Steady state efficiency
20.	FootPrint	164.0 mm ²	System	Total Foot Print Area of BOM components
21.	Frequency	541.189 kHz	System	Switching frequency
22.	Iout	500.0 mA	System	Iout operating point
23.	Mode	PFM	System	Conduction Mode
24.	Pout	1.65 W	System	Total output power

#	Name	Value	Category	Description
25.	Total BOM	\$0.9	System Information	Total BOM Cost
26.	Vin	13.0 V	System Information	Vin operating point
27.	Vout	3.3 V	System Information	Operational Output Voltage
28.	Vout Actual	3.283 V	System Information	Vout Actual calculated based on selected voltage divider resistors
29.	Vout Tolerance	2.889 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
30.	Vout p-p	16.595 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	500.0 m	Maximum Output Current
VinMax	13.0	Maximum input voltage
VinMin	11.0	Minimum input voltage
VinTyp	12.0	Typical input voltage
Vout	3.3	Output Voltage
base_pn	TPS563200	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

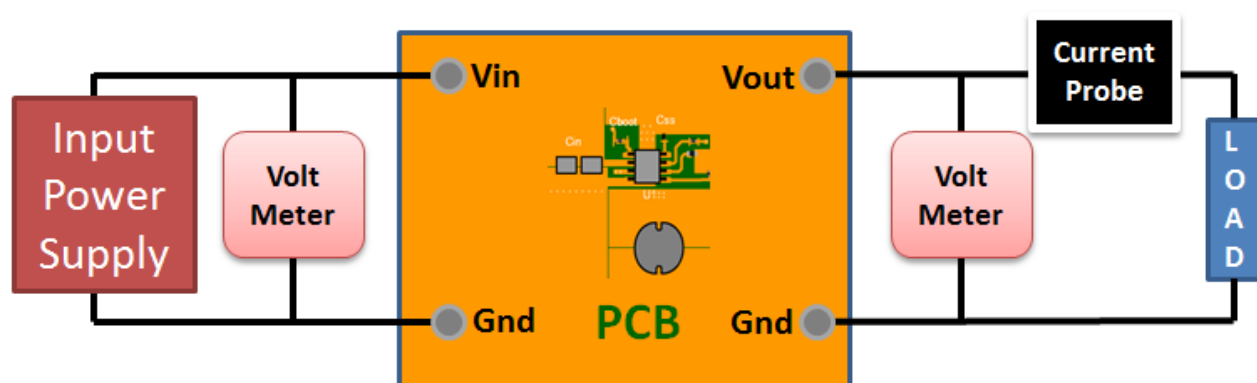
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 11.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 5FFB4611713714A73A2F511ACDE4AFEF[v1]
2. **TPS563200** Product Folder : <http://www.ti.com/product/TPS563200> : contains the data sheet and other resources.

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